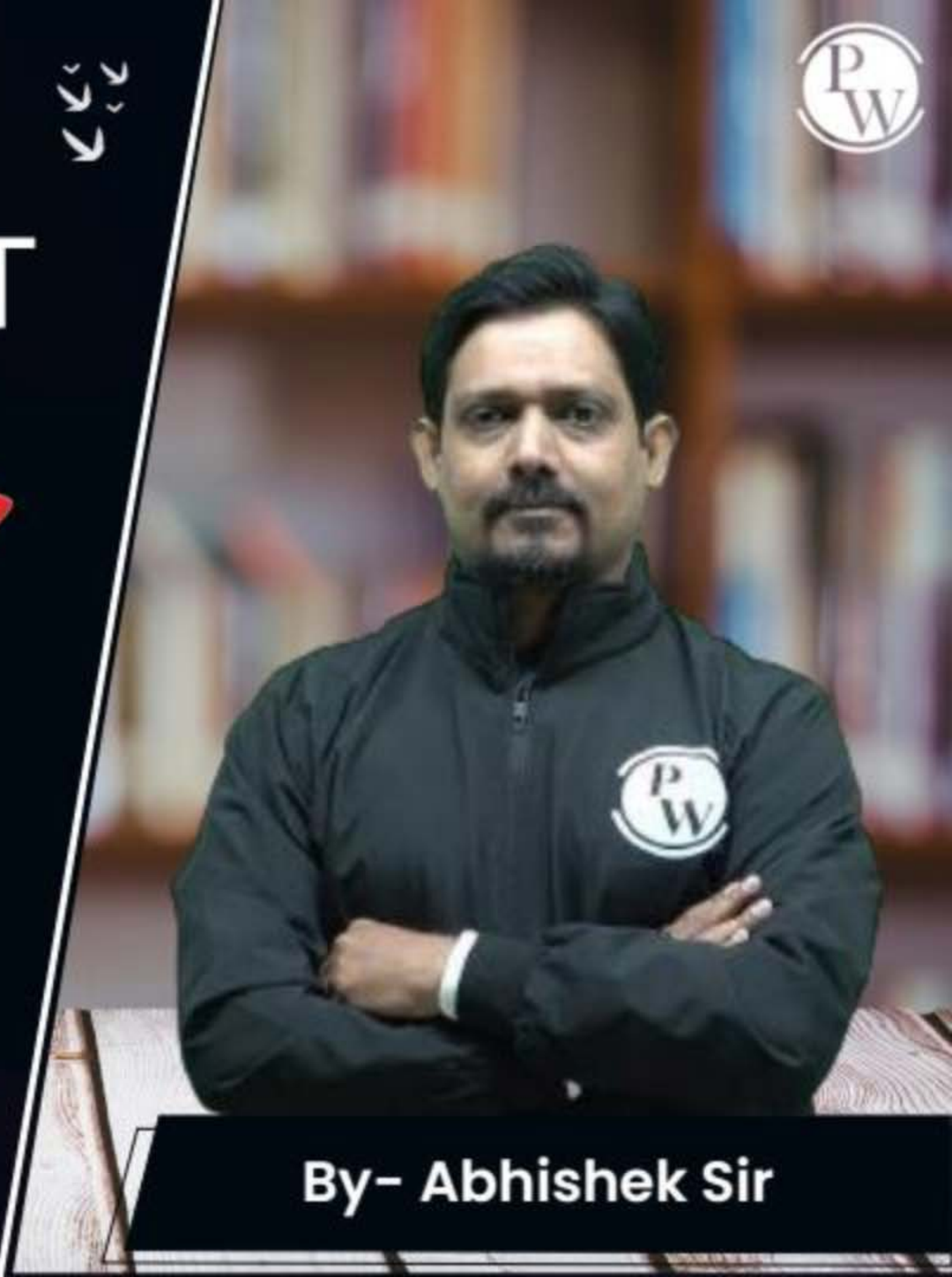
Computer Science & IT

Fundamental of C Language



Array & Pointers

Lecture No. 04



By- Abhishek Sir

Recap of Previous Lecture



Topic

Topic

Topic

Topic

Topic

1-D array

Base Address, Size of each element

formula.

Slide

Topics to be Covered



Topic 2-Dimensional Array

Topic Rows & column

Topic Memory arrangment

Topic How to access 2-D array elements

Topic

Slide

2-D Array



```
int a[3][4];
```

Rows

Columns

	Column 0	Column 1	Column 2	Column 3
Row - 0	A[0][0]	A[0][1]	A[0][2]	A[0][3]
Row - 1	A[1][0]	A[1][1]	A[1][2]	A[1][3]
Row - 2	A[2][0]	A[2][1]	A[2][2]	A[2][3]

* In 2-D array multiple 1-D array will be present
* In this declaration there are 3 1-D array present

Slide



2-D Array

```
int a1[2][3] = {{1,2,3},{4,5,6}};
```

* a_1 has 2 Rows & 3 columns
* a_1 has 2 - 1-D array present

	col-0	col-1	col-2
Raw-0	1	2	3
Raw-1	4	5	6

Slide



2-D Array

In C-language elements of 2D array are arrange Row wise one after another.

This arrangment is called Row major order

Slide



2-D Array

```
int a1[2][3] = { {1,2,3}, {4,5,6} };
```

	col-0	col-1	col-2
Row 0	1	2	3
Row 1	4	5	6

```
printf("%d", a1[0][0]);
// print 1

printf("%d", a1[0][2]);
// print 3
```

Question

If you want to print 6 then what should be the print statement?

```
printf("%d", a1[1][2]);
```

Slide



2-D Array

Memory Arrangements

Array element stored in Sequential manner

2-D Mapping 1 - D



Slide



2-D Array

Accessing of 1-D array

fragment

```
int a[10] = {1,2,3,4,5,6,7,8,9,10};
int i;
for (i=0; i<10; i++)        // loop
    printf("%d \n", a[i]);
```

Slide



2-D Array

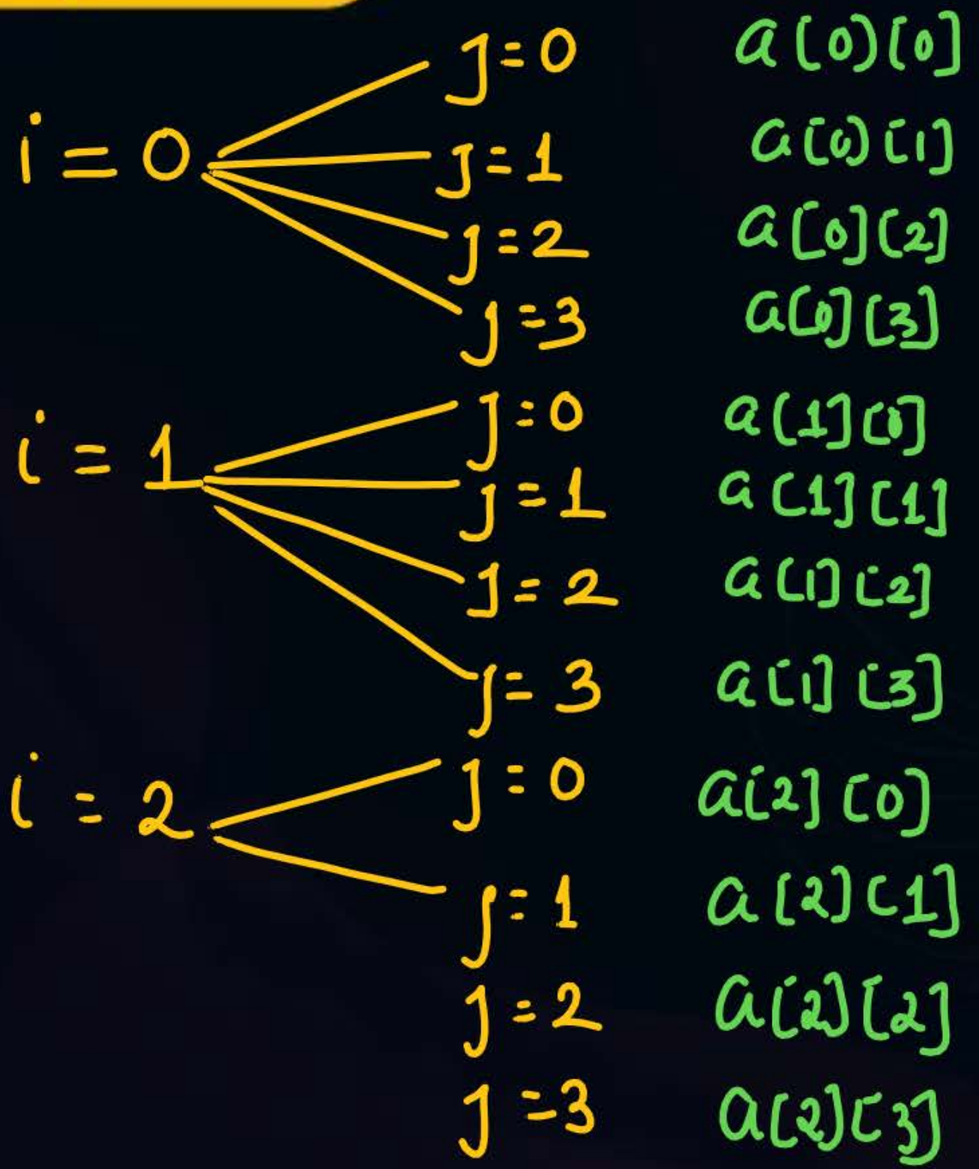
```
int a[3][4] = {{1,2,3,4},
               {5,6,7,8},
               {9,10,11,12}};
```

	col-0	col-1	col-2	col-3
Row-0	1	2	3	4
Row-1	5	6	7	8
Row-2	9	10	11	12

```
#include<stdio.h>
int main(){
int a[3][4]={{1,2,3,4},{5,6,7,8},{9,10,11,12}};
int i,j;
for(i=0;i<3;i++){
    for(j=0;j<4;j++){
        printf("%d \t",a[i][j]);
    }
    printf("\n");
}
return 0;}
```

Slide

2-D Array



Slide



2 mins Summary

Topic — 2-D array

Topic — Rows & columns

Topic — Memory arrangment

Topic — How to access 1-D & 2-D array using loop

Topic

THANK - YOU

